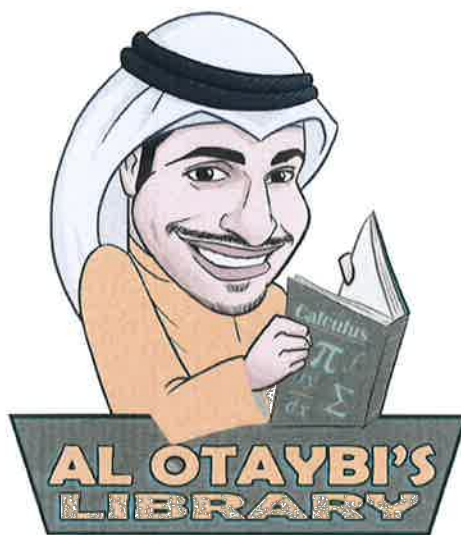


Kuwait University

2021/2022

Spring course



Calculus B

Second

Section : **11.3 – 11.4 part (a/b)**

11.3 The Integral Test and Estimates of Sums.

11.4 The Comparison Tests.

(10)



11.3 Integral test



اختبار يتم تجربته في حال كان السؤال قابل للتكامل

دالة ومشتقتها

Examples

$$\sum \frac{\tan^{-1} n}{n^2 + 1}$$

$$\sum \frac{1}{n \ln^k n}$$

عند وجود أولوية
by parts

Examples

$$\sum n e^{-n}$$

$$\sum n 2^{-n}$$

زاوية أو أس ليس حدودية
من الدرجة الأولى

Examples

$$\sum n^2 e^{-n^3}$$

شروط الحل

if $\sum_{n=1}^{\infty} a_n$

let $f(x) = a_n$



يتم استبدال كل
 $x \rightarrow n$

1) $f(x)$ is positive and continuous on $[1, \infty)$

2) $f(x)$ is decreasing on $[1, \infty)$



$$f'(x) < 0 \quad \forall x \in [1, \infty)$$

If yes

$$I = \int_1^{\infty} f(x) dx$$

= عدد

I is CONVERGENT

$$\sum_{n=1}^{\infty} a_n \text{ is CONVERGENT}$$

$= \pm \infty$ or DNE

I is DIVERGENT

$$\sum_{n=1}^{\infty} a_n \text{ is DIVERGENT}$$

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{1}{n(1+\ln n)}$$

Let $f(x) = \frac{1}{x(1+\ln x)}$

دالة ومشتقتها



1) $f(x)$ is continuous and positive on $[1, \infty)$

$$2) f'(x) = \frac{0 - (1) \left[(1)(1+\ln x) + x \cdot \frac{1}{x} \right]}{[x(1+\ln x)]^2}$$

$$= \frac{-[2+\ln x]}{[x(1+\ln x)]^2} < 0 \quad \forall x \in [1, \infty)$$

$\therefore f(x)$ is decreasing on $[1, \infty)$

$$I = \int_1^{\infty} \frac{1}{x(1+\ln x)} dx$$

$$I_1 = \int \frac{1}{x(1+\ln x)} dx$$

Let $u = 1 + \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{u} du = \ln|u| + C = \ln|1+\ln x| + C$$

$$I = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x(1+\ln x)} dx$$

$$= \lim_{t \rightarrow \infty} [\ln|1+\ln x|]_1^t$$

$$= \lim_{t \rightarrow \infty} [\ln|1+\ln t| - \ln|1+\ln 1|]$$

$$= \ln \infty - \ln 1 = \infty - 0 = \infty$$

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent (Integral test)

Determine whether the series is convergent or divergent $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^2 + 1}$

$$\hookrightarrow f(x) = \frac{\tan^{-1} x}{x^2 + 1}$$

① $\hookrightarrow f(x)$ is positive and continuous on $[1, \infty)$

$$\textcircled{2} \hookrightarrow f'(x) = \frac{\cancel{(x^2+1)} \cdot \frac{1}{\cancel{x^2+1}} - \tan^{-1} x \cdot 2x}{[x^2+1]^2}$$

$$\hookrightarrow f'(x) = \frac{1 - 2x \tan^{-1} x}{[x^2+1]^2} < 0 \quad \forall x \in [1, \infty)$$

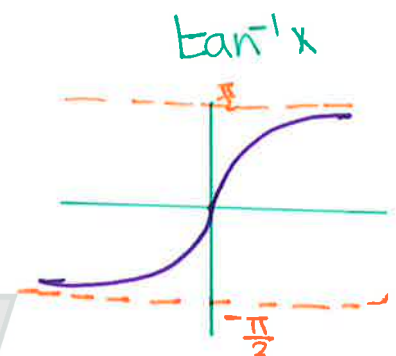
$\hookrightarrow f(x)$ is decreasing on $[1, \infty)$

$$I = \int_1^{\infty} \frac{\tan^{-1} x}{x^2 + 1} dx$$

$$I_1 = \int \frac{\tan^{-1} x}{x^2 + 1} dx$$

$$\text{Let } u = \tan^{-1} x \\ du = \frac{1}{x^2 + 1} dx$$

$$\hookrightarrow \int u du = \frac{u^2}{2} + C = \frac{(\tan^{-1} x)^2}{2} + C$$



See Next page plz!

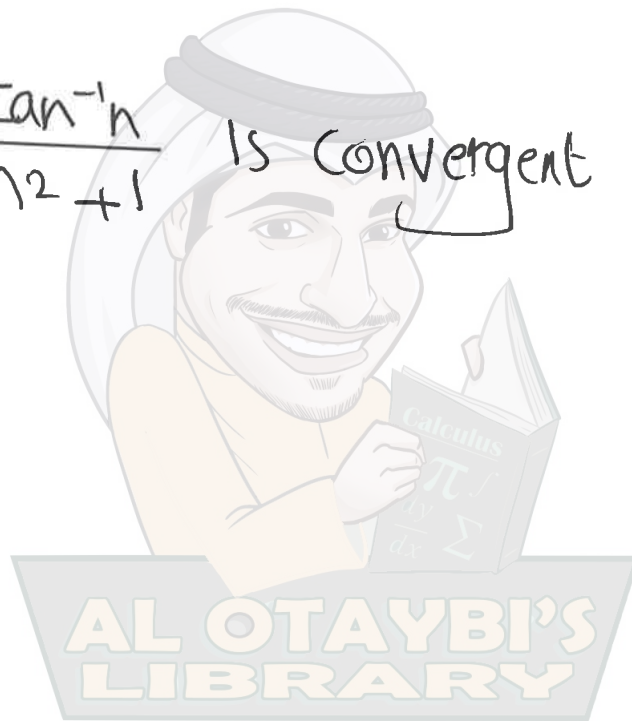
$$I = \lim_{t \rightarrow \infty} \int_1^t \frac{\tan^{-1} x}{x^2 + 1} dx$$

$$= \lim_{t \rightarrow \infty} \left[\frac{(\tan^{-1} x)^2}{2} \right]_1^t$$

$$= \frac{1}{2} \lim_{t \rightarrow \infty} [(\tan^{-1} t)^2 - (\tan^{-1} 1)^2]$$

$$= \frac{1}{2} \cdot \left[\left(\frac{\pi}{2} \right)^2 - \left(\frac{\pi}{4} \right)^2 \right] \Rightarrow \text{Convergent}$$

∴ $\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^2 + 1}$ is Convergent "Integral test"



Determine whether the series is convergent or divergent

$$\sum_{n=2}^{\infty} \frac{1}{n \ln(n^2)} \rightarrow a_n$$

Dec 2017

$$\sum_{n=2}^{\infty} \frac{1}{n \cdot 2 \ln n} = \sum_{n=2}^{\infty} \frac{1}{2n \ln n}$$

Let $f(x) = \frac{1}{2x \ln x}$

1) $f(x)$ is continuous and positive on $[2, \infty)$

$$\begin{aligned} 2) f'(x) &= \frac{0 - (1) \left[2(1) \ln x + 2x \cdot \frac{1}{x} \right]}{(2x \ln x)^2} \\ &= \frac{-[2 \ln x + 2]}{(2x \ln x)^2} < 0 \quad \forall x \in [2, \infty) \end{aligned}$$

$\therefore f(x)$ is decreasing on $[2, \infty)$

$$I = \int_2^{\infty} \frac{1}{2x \ln x} dx$$

$$I_1 = \int \frac{1}{2x \ln x} dx$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{2u} du = \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|\ln x| + C$$

$$I = \lim_{t \rightarrow \infty} \int_2^t \frac{1}{2x \ln x} dx$$

$$\begin{aligned} &= \lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln|\ln x| \right]_2^t = \frac{1}{2} \lim_{t \rightarrow \infty} [\ln|\ln t| - \ln|\ln 2|] \\ &= \frac{1}{2} [\ln \infty - \ln \ln 2] = \infty \end{aligned}$$

$\therefore \sum_{n=2}^{\infty} a_n$ is divergent (Integral test)

Determine whether the series is convergent or divergent . $\sum_{n=2}^{\infty} \frac{1}{n \ln^2 n}$

$$f(x) = \frac{1}{x \ln^2 x}$$

① $f(x)$ is positive and continuous on $[2, \infty)$

$$\begin{aligned} \textcircled{2} \quad f'(x) &= \frac{x \cdot \ln^2 x \cdot (0) - 1 \cdot [(1) \ln^2 x + x \cdot 2 \ln x \cdot \frac{1}{x}]}{[x \ln^2 x]^2} \\ &= \frac{- [\ln^2 x + 2 \ln x]}{[x \ln^2 x]^2} < 0 \quad \forall x \in [2, \infty) \end{aligned}$$

$f(x)$ is decreasing on $[2, \infty)$

$$I = \int_2^{\infty} \frac{1}{x \ln^2 x} dx$$

$$I_1 = \int \frac{1}{x (\ln x)^2} dx$$

$$\begin{aligned} \text{Let } u &= \ln x \\ du &= \frac{1}{x} dx \end{aligned}$$

$$\int \frac{1}{u^2} du$$

$$\int u^{-2} du = \frac{u^{-1}}{-1} + c = -\frac{1}{u} + c = -\frac{1}{\ln x} + c$$

See next page plz!!

$$I = \lim_{t \rightarrow \infty} \int_2^t \frac{1}{x \ln^2 x} dx$$

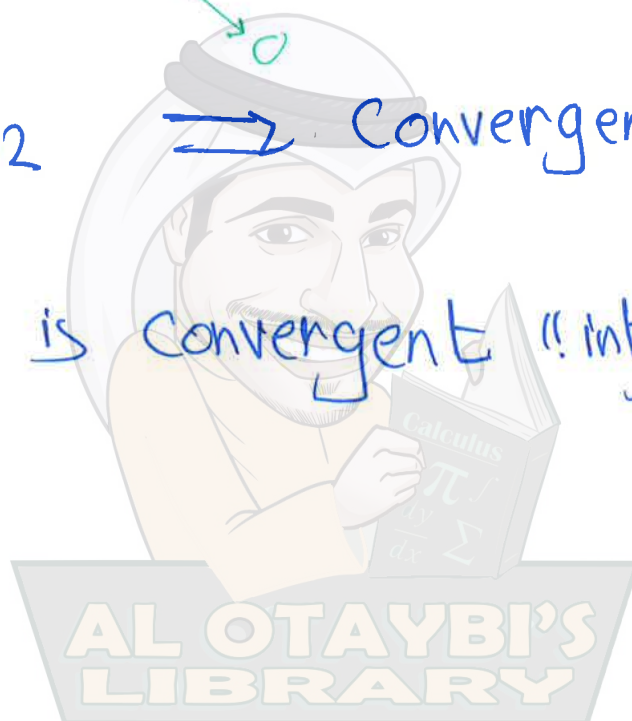
$$= \lim_{t \rightarrow \infty} \left[\frac{-1}{\ln x} \right]_2^t$$

$$= - \lim_{t \rightarrow \infty} \left[\frac{1}{\ln t} - \frac{1}{\ln 2} \right]$$

$$= - \left[\frac{1}{\infty} - \frac{1}{\ln 2} \right]$$

$$= \frac{1}{\ln 2} \Rightarrow \text{Convergent}$$

$\therefore \sum_{n=2}^{\infty} a_n$ is convergent "integral test"



Determine whether the series is convergent or divergent.

$$\sum_{n=1}^{\infty} n^2 e^{-n^3}$$

$$f(x) = x^2 e^{-x^3}$$

① $f(x)$ is positive and continuous on $[1, \infty)$

$$② f'(x) = 2x e^{-x^3} + x^2 e^{-x^3} \cdot (-3x^2)$$

$$f'(x) = 2x e^{-x^3} - 3x^4 e^{-x^3}$$

$$f'(x) = x e^{-x^3} [2 - 3x^3] < 0 \quad \forall x \in [1, \infty)$$

$f(x)$ is decreasing on $[1, \infty)$

$$I = \int_1^{\infty} x^2 e^{-x^3} dx$$

$$I_1 = \int x^2 e^{-x^3} dx$$

$$\text{Let } u = x^3$$

$$du = 3x^2 dx \rightarrow \frac{1}{3} du = x^2 dx$$

$$= \frac{1}{3} \int e^{-u} du$$

$$= \frac{1}{3} \frac{e^{-u}}{-1} + c = -\frac{1}{3} e^{-x^3} + c$$

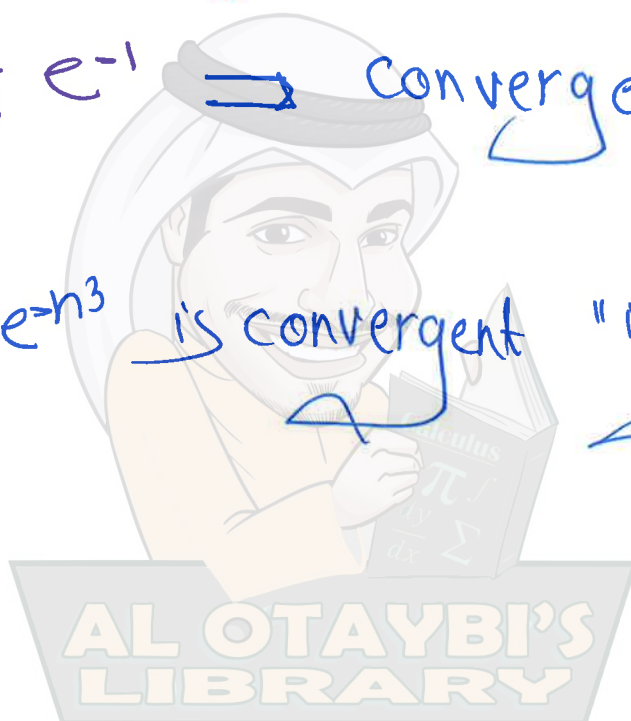
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$$\begin{aligned}
 I &= \lim_{t \rightarrow \infty} \int_1^t x^2 e^{-x^3} dx \\
 &= \lim_{t \rightarrow \infty} \left[\frac{-1}{3} e^{-x^3} \right]_1^t \\
 &= \frac{-1}{3} \lim_{t \rightarrow \infty} [e^{-t^3} - e^{-1}]
 \end{aligned}$$

$$= \frac{-1}{3} [e^{-\infty} - e^{-1}]$$

$$= \frac{1}{3} e^{-1} \Rightarrow \text{convergent}$$

$\therefore \sum_{n=1}^{\infty} n^2 e^{-n^3}$ is convergent "integral test"



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n} e^{\sqrt{n}}}$$

$$f(x) = \frac{1}{\sqrt{x} e^{\sqrt{x}}}$$

① $f(x)$ is positive and continuous on $[1, \infty)$

$$f'(x) = \frac{\sqrt{x} e^{\sqrt{x}} (0) - 1 \cdot \left[\frac{1}{2\sqrt{x}} e^{\sqrt{x}} + \sqrt{x} e^{\sqrt{x}} \frac{1}{2\sqrt{x}} \right]}{[\sqrt{x} e^{\sqrt{x}}]^2}$$

$$f'(x) = \frac{- \left[\frac{1}{2\sqrt{x}} e^{\sqrt{x}} + \frac{1}{2} e^{\sqrt{x}} \right]}{[\sqrt{x} e^{\sqrt{x}}]^2} < 0 \quad \forall x \in [1, \infty)$$

$f(x)$ is decreasing on $[1, \infty)$

$$I = \int_1^{\infty} \frac{1}{\sqrt{x} e^{\sqrt{x}}} dx$$

$$I_1 = \int \frac{1}{\sqrt{x} e^{\sqrt{x}}} dx$$

$$\text{Let } u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}} dx \rightarrow 2du = \frac{1}{\sqrt{x}} dx$$

$$\begin{aligned} I_1 &= 2 \int \frac{1}{e^u} du = 2 \int e^{-u} du = 2 \frac{e^{-u}}{-1} + C \\ &= -2e^{-\sqrt{x}} + C \end{aligned}$$

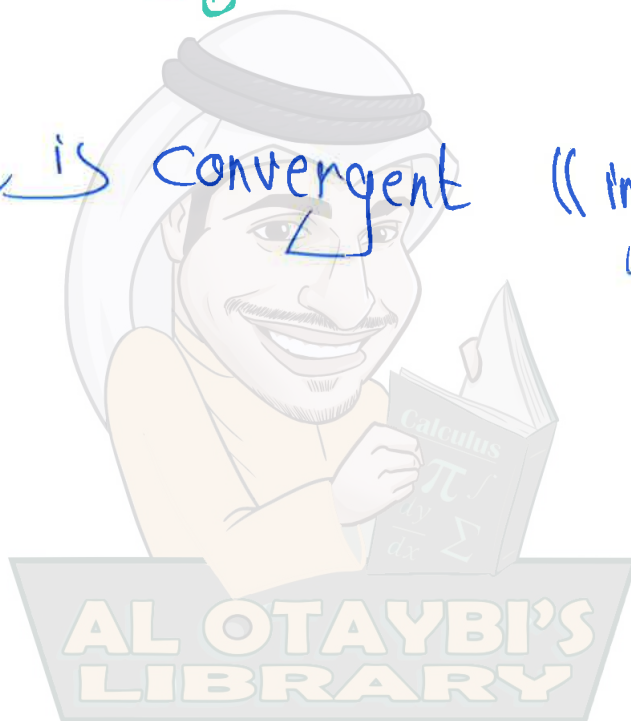
$$I = \lim_{t \rightarrow \infty} \int_1^t x^2 e^{-x^3} dx$$

$$= \lim_{t \rightarrow \infty} [-2e^{-\sqrt{x}}]_1^t$$

$$= -2 \lim_{t \rightarrow \infty} [e^{-\sqrt{t}} - e^{-1}]$$

$$= -2 [e^{-\infty} - e^{-1}] = 2e^{-1} \Rightarrow \text{Convergent}$$

∴ $\sum a_n$ is convergent (Integral test)



Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} ne^{-n} \longrightarrow a_n$$

Let $f(x) = xe^{-x}$

1) $f(x)$ is continuous and positive on $[1, \infty)$

2) $f'(x) = (1)e^{-x} + xe^{-x}(-1)$

$$= e^{-x}[1-x] < 0 \quad \forall x \in (1, \infty)$$

$\therefore f(x)$ is decreasing on $(1, \infty)$

$$I = \int_1^{\infty} xe^{-x} dx$$

$$I_1 = \int xe^{-x} dx$$

$$u = x \quad dv = e^{-x} dx$$

$$du = dx \quad v = -e^{-x}$$

$$-xe^{-x} + \int e^{-x} dx$$

$$-xe^{-x} - e^{-x} + C$$

$$\frac{-x}{e^x} - e^{-x} + C$$

$$I = \lim_{t \rightarrow \infty} \int_1^t xe^{-x} dx$$

$$= \lim_{t \rightarrow \infty} \left[\frac{-x}{e^x} - e^{-x} \right]_1^t$$

$$\lim_{t \rightarrow \infty} \frac{-t}{e^t} = \lim_{t \rightarrow \infty} \left[\left(\frac{-t}{e^t} - e^{-t} \right) - \left(\frac{-1}{e^1} - e^{-1} \right) \right]$$

L'H $\lim_{t \rightarrow \infty} \frac{-1}{e^t} = (0 - 0) - \left(\frac{-1}{e} - e^{-1} \right) = - \left(\frac{-1}{e} - \frac{1}{e} \right) = - \frac{-2}{e} = \frac{2}{e}$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Integral test)

Determine whether the following series is convergent or divergent

May 2017

$$\sum_{n=2}^{\infty} \frac{\ln n}{n^2}$$

$$\rightarrow \sum_{n=2}^{\infty} a_n$$

Let $f(x) = \frac{\ln x}{x^2}$

1) $f(x)$ is continuous and positive on $[2, \infty)$

$$\begin{aligned} 2) f'(x) &= \frac{x^2 \cdot \frac{1}{x} - \ln x \cdot 2x}{x^4} = \frac{x - 2x \ln x}{x^4} \\ &= \frac{x(1 - 2 \ln x)}{x^4} = \frac{1 - 2 \ln x}{x^3} < 0 \quad \forall x \in [2, \infty) \end{aligned}$$

$\Rightarrow \therefore f(x)$ is decreasing on $[2, \infty)$

$$I = \int_2^{\infty} \frac{\ln x}{x^2} dx$$

$$I = \int \frac{\ln x}{x^2} dx$$

$$u = \ln x$$

$$dv = \frac{1}{x^2} dx$$

$$du = \frac{1}{x} dx$$

$$v = \frac{x^{-1}}{-1}$$

$$- \ln x \cdot x^{-1} + \int x^{-1} \cdot \frac{1}{x} dx$$

$$- \frac{\ln x}{x} + \frac{x^{-1}}{-1} + C$$

$$- \frac{\ln x}{x} - \frac{1}{x} + C$$

$$\lim_{t \rightarrow \infty} \int_2^t \frac{\ln x}{x^2} dx$$

$$\lim_{t \rightarrow \infty} \left[\frac{-\ln x}{x} - \frac{1}{x} \right]_2^t$$

$$\lim_{t \rightarrow \infty} \left[\left(\frac{-\ln t}{t} - \frac{1}{t} \right) - \left(\frac{-\ln 2}{2} - \frac{1}{2} \right) \right]$$

$$0 - 0 - \left(\frac{-\ln 2}{2} - \frac{1}{2} \right) = \frac{\ln 2}{2} + \frac{1}{2}$$

$\lim_{t \rightarrow \infty} \frac{-\ln t}{t}$
L'Hop $\lim_{t \rightarrow \infty} \frac{-\frac{1}{t}}{1} = \lim_{t \rightarrow \infty} -\frac{1}{t} = -\frac{1}{\infty} = 0$

$\therefore \sum_{n=2}^{\infty} a_n$ is convergent (Integral test)

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Text book question



Determine whether the series is convergent or divergent.

$$\sum_{n=3}^{\infty} \frac{n^2}{e^n}$$

Let $f(x) = \frac{x^2}{e^x}$

1) $f(x)$ is continuous and positive on $[3, \infty)$

$$\begin{aligned} 2) f'(x) &= \frac{e^x \cdot 2x - x^2 e^x (1)}{(e^x)^2} = \frac{x e^x [2 - x]}{(e^x)^2} \\ &= \frac{x [2 - x]}{e^x} < 0 \quad \forall x \in [3, \infty) \end{aligned}$$

$\therefore f(x)$ is decreasing on $[3, \infty)$

$$I = \int_3^{\infty} \frac{x^2}{e^x} dx$$

$$I_1 = \int x^2 e^{-x} dx$$

$$u = x^2 \quad dv = e^{-x} dx$$

$$du = 2x dx \quad v = -e^{-x}$$

$$-x^2 e^{-x} + 2 \int x e^{-x} dx$$

$$u = x \quad dv = e^{-x} dx$$
$$du = dx \quad v = -e^{-x}$$

$$-x e^{-x} + \int e^{-x} dx$$

$$-x e^{-x} - e^{-x} + c$$

$$\Rightarrow -\frac{x^2}{e^x} + 2 \left[\frac{-x}{e^x} - \frac{1}{e^x} \right] + c$$

$$\begin{aligned}
 \therefore I &= \lim_{t \rightarrow \infty} \int_3^t \frac{x^2}{e^x} dx \\
 &= \lim_{t \rightarrow \infty} \left[\frac{-x^2}{e^x} + 2 \left[\frac{-x}{e^x} - \frac{1}{e^x} \right] \right]_3^t \\
 &= \lim_{t \rightarrow \infty} \left[\left(\frac{-t^2}{e^t} + 2 \left[\frac{-t}{e^t} - \frac{1}{e^t} \right] \right) - \left(\frac{-9}{e^3} + 2 \left[\frac{-3}{e^3} - \frac{1}{e^3} \right] \right) \right] \\
 &= L
 \end{aligned}$$

$$\lim_{t \rightarrow \infty} \frac{-t^2}{e^t}$$

L.H

$$\lim_{t \rightarrow \infty} \frac{-2t}{e^t}$$

L.H

$$\lim_{t \rightarrow \infty} \frac{-2}{e^t} = \frac{-2}{\infty} = 0$$

$$\lim_{t \rightarrow \infty} \frac{-t}{e^t}$$

L.H

$$\lim_{t \rightarrow \infty} \frac{-1}{e^t}$$

$$= \frac{-1}{\infty} = 0$$

$$\therefore L = (0 + 2[0 - 0]) - \left(\frac{-9}{e^3} + 2 \left[\frac{-3}{e^3} - \frac{1}{e^3} \right] \right)$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Integral test)

(a) State the integral test

(b) Use the integral test to determine whether the series

$$\sum_{n=0}^{\infty} \frac{2^n}{1+4^n}$$

(a)

The Integral Test Suppose f is a continuous, positive, decreasing function on $[1, \infty)$ and let $a_n = f(n)$. Then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if the improper integral $\int_1^{\infty} f(x) dx$ is convergent. In other words:

(i) If $\int_1^{\infty} f(x) dx$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is convergent.

(ii) If $\int_1^{\infty} f(x) dx$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Let $f(x) = \frac{2^x}{1+4^x}$

(1) $f(x)$ is continuous and positive on $[0, \infty)$

(2) $f'(x) = \frac{(1+4^x)2^x \ln 2 - 2^x 4^x \ln 4}{(1+4^x)^2}$

$$= \frac{2^x [(1+4^x) \ln 2 - 4^x \ln 4]}{(1+4^x)^2} = \frac{2^x [\ln 2 + 4^x \ln 2 - 4^x \ln 4]}{(1+4^x)^2}$$

$$= \frac{2^x [\ln 2 + 4^x (\ln 2 - \ln 4)]}{(1+4^x)^2} < 0 \quad \forall x \in [0, \infty)$$

$\Rightarrow \therefore f(x)$ is decreasing on $[0, \infty)$

$$I = \int_0^{\infty} \frac{2^x}{1+(2^x)^2} dx$$

← See Next page!

$$I_1 = \int \frac{2^x}{1 + (2^x)^2} dx$$

Let $u = 2^x$

$$du = 2^x \ln 2 dx \longrightarrow \frac{1}{\ln 2} du = 2^x dx$$

$$\frac{1}{\ln 2} \int \frac{1}{1 + u^2} du$$

$$\frac{1}{\ln 2} \tan^{-1} u + C = \frac{1}{\ln 2} \tan^{-1}(2^x) + C$$

$$\lim_{t \rightarrow \infty} \int_0^t \frac{2^x}{1 + (2^x)^2} dx$$

$$\lim_{t \rightarrow \infty} \left[\frac{1}{\ln 2} \tan^{-1}(2^x) \right]_0^t$$

$$\frac{1}{\ln 2} \lim_{t \rightarrow \infty} [\tan^{-1}(2^t) - \tan^{-1}(2^0)]$$

$$\frac{1}{\ln 2} [\tan^{-1} \infty - \tan^{-1}(1)]$$

$$\frac{1}{\ln 2} \left[\frac{\pi}{2} - \frac{\pi}{4} \right] = \frac{1}{\ln 2} \left[\frac{2\pi}{4} - \frac{\pi}{4} \right] = \frac{1}{\ln 2} \frac{\pi}{4}$$

$\Rightarrow \therefore \sum_{n=0}^{\infty} a_n$ is convergent (Integral test)



Home work



(a) State the integral test

(b) Use the integral test to determine whether the series

$$\sum_{n=1}^{\infty} \frac{1}{n^2 + 6n + 13}$$



Text book question



Find the values of p for which the series is convergent.

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^p} \rightarrow a_n$$

Let $f(x) = \frac{1}{x(\ln x)^p}$

1) $f(x)$ is continuous and positive on $[2, \infty)$

$$\begin{aligned} 2) f'(x) &= \frac{0 - (1) \left[(1)(\ln x)^p - x \cdot p(\ln x)^{p-1} \cdot \frac{1}{x} \right]}{[x(\ln x)^p]^2} \\ &= \frac{-[(\ln x)^p - p(\ln x)^{p-1}]}{[x(\ln x)^p]^2} < 0 \quad \forall x \in [2, \infty) \end{aligned}$$

$\therefore f(x)$ is decreasing on $[2, \infty)$

$$I = \int_2^{\infty} \frac{1}{x(\ln x)^p} dx$$

$$I_1 = \int \frac{1}{x(\ln x)^p} dx$$

Let $u = \ln x$
 $du = \frac{1}{x} dx$

$$\int \frac{1}{u^p} du = \int u^{-p} du = \frac{u^{-p+1}}{-p+1} + C$$

$$= \frac{(\ln x)^{-p+1}}{-p+1} + C$$

$$\lim_{t \rightarrow \infty} \int_2^t \frac{1}{x(\ln x)^p} dx$$

$$\lim_{t \rightarrow \infty} \left[\frac{(\ln x)^{-p+1}}{-p+1} \right]_2^t = \frac{1}{-p+1} \lim_{t \rightarrow \infty} [(\ln t)^{-p+1} - (\ln 2)^{-p+1}]$$

$\therefore \sum_{n=2}^{\infty} a_n$ is convergent when $-p+1 < 0 \rightarrow 1 < p \rightarrow p \in (1, \infty)$

Show that the series is convergent using integral test

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

Let $f(x) = \frac{1}{x^3} = x^{-3}$

لو لم يحدد الطريقة يتم حلها

P-series



1) $f(x)$ is continuous and positive on $[1, \infty)$

2) $f'(x) = -3x^{-4} = \frac{-3}{x^4} < 0 \quad \forall x \in [1, \infty)$

$\therefore f(x)$ is decreasing on $[1, \infty)$

$$I = \int_1^{\infty} \frac{1}{x^3} dx$$

$$I_1 = \int \frac{1}{x^3} dx$$

$$= \int x^{-3} dx = \frac{x^{-2}}{-2} + C = -\frac{1}{2} \frac{1}{x^2} + C$$

$$I = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^3} dx$$

$$= \lim_{t \rightarrow \infty} \left[-\frac{1}{2} \frac{1}{x^2} \right]_1^t$$

$$= -\frac{1}{2} \lim_{t \rightarrow \infty} \left[\frac{1}{t^2} - \frac{1}{1} \right]$$

$$= -\frac{1}{2} \left[0 - \frac{1}{1} \right] = \frac{1}{2}$$

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (Integral test)



11.4 The comparison Test



Basic Comparison Test
(B.C.T)

Limit Comparison Test
(L.C.T)



Basic Comparison Test (B.C.T)



Suppose that

$$\sum a_n$$

and

$$\sum b_n$$

are series with positive terms

If $a_n \leq b_n$ and

$$\sum b_n$$

is convergent



$\sum a_n$ is convergent

If $a_n \geq b_n$ and

$$\sum b_n$$

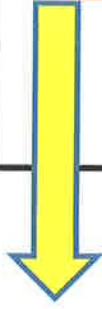
is divergent



$\sum a_n$ is divergent

طريقة الحل

- (1) ربط السؤال بسيريس آخر عن طريق علاقة
- (2) حل السيريس الآخر بأحد الطرق الأخرى
- (3) تطبيق النظرية



العلاقات



$\sin, \cos$

$$-1 \leq \sin^k n \leq 1$$

$$-1 \leq \cos^k n \leq 1$$

k is of odd

$$0 \leq \sin^k n \leq 1$$

$$0 \leq \cos^k n \leq 1$$

k is of even

المقام +

$$\frac{1}{a+b} < \frac{1}{a}$$

$$\frac{1}{a-b} > \frac{1}{a}$$

$\ln$

$$\ln(n) < n$$

$$\forall n > 0$$

$$\ln(n) > 1$$

$$\forall n \geq 3$$

$\tan^{-1}$

$$\frac{\pi}{4} \leq \tan^{-1} n < \frac{\pi}{2}$$

$$\forall n \geq 1$$

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\cos(n)}{\sqrt{n^3+1}}$$

$$-1 \leq \cos(n) \leq 1$$

$$\div \sqrt{n^3+1}$$

$$\frac{-1}{\sqrt{n^3+1}} \leq \frac{\cos(n)}{\sqrt{n^3+1}} \leq \frac{1}{\sqrt{n^3+1}}$$

$$\lim_{n \rightarrow \infty} \frac{-1}{\sqrt{n^3+1}} = \frac{-1}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^3+1}} = \frac{1}{\infty} = 0$$

∴ By Sandwich Thm $\Rightarrow \lim_{n \rightarrow \infty} \frac{\cos(n)}{\sqrt{n^3+1}} = 0$

~~G.T.N.P.X~~

N.B

مُرب

مثلاً الاختبار

$$-1 \leq \cos(n) \leq 1$$

$$\div \sqrt{n^3+1}$$

$$\frac{-1}{\sqrt{n^3+1}} \leq \frac{\cos(n)}{\sqrt{n^3+1}} \leq \frac{1}{\sqrt{n^3+1}}$$

$$\frac{\cos(n)}{\sqrt{n^3+1}} \leq \frac{1}{\sqrt{n^3+1}} < \frac{1}{\sqrt{n^3}}$$

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^3}} = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$$

$P = \frac{3}{2} > 1$

∴ $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is convergent (P-series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\cos n}{\sqrt{n^3+1}}$ is convergent (B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=2}^{\infty} \frac{1 + \sin^2 n}{\sqrt{n^2 - 1}}$$

$$0 \leq \sin^2 n \leq 1$$

(+1)

$$1 \leq 1 + \sin^2 n \leq 2$$

$$\div \sqrt{n^2 - 1}$$

$$\frac{1}{\sqrt{n^2 - 1}} \leq \frac{1 + \sin^2 n}{\sqrt{n^2 - 1}} \leq \frac{2}{\sqrt{n^2 - 1}}$$

$$\frac{1}{\sqrt{n^2}} \leq \frac{1}{\sqrt{n^2 - 1}} \leq \frac{1 + \sin^2 n}{\sqrt{n^2 - 1}}$$

$\rightarrow \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent (Harmonic)

$$\therefore \sum_{n=2}^{\infty} \frac{1 + \sin^2 n}{\sqrt{n^2 - 1}} \text{ is divergent (B.C.T.)}$$

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\sin(\sqrt{n})}{\sqrt[3]{n^7 + 3}}$$

$$-1 \leq \sin(\sqrt{n}) \leq 1$$

$$\div \sqrt[3]{n^7 + 3}$$

$$\frac{-1}{\sqrt[3]{n^7 + 3}} \leq \frac{\sin(\sqrt{n})}{\sqrt[3]{n^7 + 3}} \leq \frac{1}{\sqrt[3]{n^7 + 3}} < \frac{1}{\sqrt[3]{n^7}}$$

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n^7}}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{7}{3}}}$$

$$p = \frac{7}{3} > 1$$

is convergent (p-series)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{2 - \cos(n)}{\sqrt{n-1}}$$

$$-1 \leq \cos(n) \leq 1$$

$$\times -1$$

$$1 \geq -\cos(n) \geq -1$$

$$+2$$

$$3 \geq 2 - \cos(n) \geq 1$$

$$\div \sqrt{n-1}$$

$$\frac{3}{\sqrt{n-1}} \geq \frac{2 - \cos(n)}{\sqrt{n-1}} \geq \frac{1}{\sqrt{n-1}} \geq \frac{1}{\sqrt{n}}$$

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad P = \frac{1}{2} < 1$$

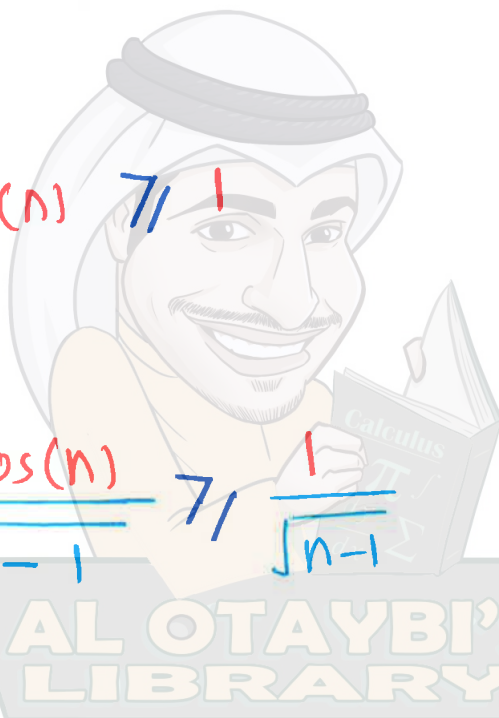
$\therefore \sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{2}}}$ is divergent
P-series

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{2 - \cos(n)}{\sqrt{n-1}}$ is divergent (B.C.T)

~~G.T.H.P.~~

~~N.B~~

مبين راج تفشل



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Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{3 - \cos(n)}{n^{\frac{2}{3}} - 2}$$

$$-1 \leq \cos(n) \leq 1$$

$$\Rightarrow -1$$

$$1 \geq -\cos(n) \geq -1$$

$$+3$$

$$4 \geq 3 - \cos(n) \geq 2$$

$$\div (n^{\frac{2}{3}} - 2)$$

$$\frac{4}{n^{\frac{2}{3}} - 2} \geq \frac{3 - \cos(n)}{n^{\frac{2}{3}} - 2} \geq \frac{2}{n^{\frac{2}{3}} - 2}$$

$$\geq \frac{2}{n^{\frac{2}{3}}}$$

$$\sum_{n=1}^{\infty} \frac{2}{n^{\frac{2}{3}}} \quad p = \frac{2}{3} < 1$$

$\therefore \sum_{n=1}^{\infty} \frac{2}{n^{\frac{2}{3}}}$ is divergent (P-series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{3 - \cos(n)}{n^{\frac{2}{3}} - 2}$ is divergent (B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\sin(7n)}{7^n + n^7}$$

Solution # 1

Solution #2

$$-1 \leq \sin(7n) \leq 1$$

$$\div (7^n + n^7)$$

$$\frac{-1}{7^n + n^7} \leq \frac{\sin(7n)}{7^n + n^7} \leq \frac{1}{7^n + n^7}$$

$$\sum_{n=1}^{\infty} \frac{1}{7^n}$$

$$\sum_{n=1}^{\infty} \left(\frac{1}{7}\right)^n$$

$$r = \frac{1}{7}$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{1}{7}\right)^n$ is convergent (Geometric Series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\sin(7n)}{7^n + n^7}$ is convergent (B.C.T)

$$-1 \leq \sin(7n) \leq 1$$

$$\div (7^n + n^7)$$

$$\frac{-1}{7^n + n^7} \leq \frac{\sin(7n)}{7^n + n^7} \leq \frac{1}{7^n + n^7}$$

$$\leq \frac{1}{n^7}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^7}$$

$$p = 7 > 1$$

$\therefore \sum_{n=1}^{\infty} \frac{1}{n^7}$ is convergent (p-series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{\sin(7n)}{7^n + n^7}$ is convergent (B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{e^n + \sin(n)}{4 + 3^n}$$

$$\begin{aligned} & -1 \leq \sin(n) \leq 1 \\ & + e^n \\ & e^n - 1 \leq e^n + \sin(n) \leq e^n + 1 \\ & \div (4 + 3^n) \end{aligned}$$

$$\frac{e^n - 1}{4 + 3^n} \leq \frac{e^n + \sin(n)}{4 + 3^n} \leq \frac{e^n + 1}{4 + 3^n} < \frac{e^n + 1}{3^n}$$

$$\sum_{n=1}^{\infty} \frac{e^n + 1}{3^n} = \sum_{n=1}^{\infty} \left[\frac{e^n}{3^n} + \frac{1}{3^n} \right]$$

$$\sum_{n=1}^{\infty} \left(\frac{e}{3} \right)^n + \sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n$$

$$r = \frac{e}{3}$$

$$|r| < 1$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{e}{3} \right)^n$ is convergent
(Geometric Series)

$$r = \frac{1}{3}$$

$$|r| < 1$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n$ is convergent
(Geometric Series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{e^n + 1}{3^n}$ is convergent

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{e^n + \sin(n)}{4 + 3^n}$ is convergent (B.C.T)

Determine whether the series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\sin^2(n) + 3^n}{n^2 + 6^n} \rightarrow \sum_{n=1}^{\infty} a_n$$

$$0 \leq \sin^2(n) \leq 1$$

$$+ 3^n$$

$$3^n \leq \sin^2(n) + 3^n \leq 1 + 3^n$$

$$\div (n^2 + 6^n)$$

$$\frac{3^n}{n^2 + 6^n} \leq \frac{\sin^2(n) + 3^n}{n^2 + 6^n} \leq \frac{1 + 3^n}{n^2 + 6^n} \leq \frac{1 + 3^n}{6^n}$$

$$\sum_{n=1}^{\infty} \left(\frac{1}{6^n} + \frac{3^n}{6^n} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{6} \right)^n + \sum_{n=1}^{\infty} \left(\frac{3}{6} \right)^n$$

$$r = \frac{1}{6}$$

$$|r| < 1$$

$\Rightarrow \therefore$ Convergent
(Geometric Series)

$$r = \frac{3}{6}$$

$$r = \frac{1}{2}$$

$$|r| < 1$$

$\Rightarrow \therefore$ Convergent
(Geometric Series)

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1 + 3^n}{6^n} \text{ is convergent}$$

$$\Rightarrow \therefore \sum_{n=1}^{\infty} a_n \text{ is convergent (B.C.T)}$$

Determine whether the following series is convergent or divergent

$$\sum_{n=2}^{\infty} \frac{1}{n \ln^2 n + 5}$$

لو 5 مو موجودة

جان ضبطت

Integration test



$$\frac{1}{n \ln^2 n + 5} < \frac{1}{n \ln^2 n}$$

$$\sum_{n=2}^{\infty} \frac{1}{n \ln^2 n}$$

$$f(x) = \frac{1}{x \ln^2 x}$$

1) $f(x)$ is continuous and positive on $[1, \infty)$

$$2) f'(x) = \frac{(x \ln^2 x)' \cdot 0 - 1 \cdot \left[(1) \ln^2 x + x \cdot 2 \ln x \cdot \frac{1}{x} \right]}{[x \ln^2 x]^2}$$

$$= \frac{-\ln x [\ln x + 2]}{x^2 \ln^4 x} = \frac{-[\ln x + 2]}{x^2 \ln^3 x} < 0 \quad \forall x \in [2, \infty)$$

$\therefore f(x)$ is decreasing on $[2, \infty)$

$$\int_2^{\infty} \frac{1}{x \ln^2 x} dx$$

$$I_1 = \int \frac{1}{x \ln^2 x} dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$= \int \frac{1}{u^2} du$$

$$= \int u^{-2} du$$

$$= \frac{u^{-1}}{-1} + C = -\frac{1}{u} + C = -\frac{1}{\ln x} + C$$

$$\lim_{t \rightarrow \infty} \int_2^t \frac{1}{x \ln^2 x} dx$$

$$\lim_{t \rightarrow \infty} \left[\frac{-1}{\ln x} \right]_2^t$$

$$\lim_{t \rightarrow \infty} \left[\frac{-1}{\ln t} + \frac{1}{\ln 2} \right]$$

$$= \frac{-1}{\infty} + \frac{1}{\ln 2}$$

$$= 0 + \frac{1}{\ln 2} = \frac{1}{\ln 2}$$

$$\therefore \sum_{n=2}^{\infty} \frac{1}{n \ln^2 n} \text{ is convergent (integral test)}$$

$$\Rightarrow \therefore \sum_{n=2}^{\infty} \frac{1}{n \ln^2 n + 5} \text{ is convergent (B.C.T)}$$

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{2 + 2^n}{3 + 3^n}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{2 + 2^n}{3 + 3^n} \\ \div 3^n \\ \lim_{n \rightarrow \infty} \frac{\frac{2}{3^n} + \frac{2^n}{3^n}}{\frac{3}{3^n} + \frac{3^n}{3^n}} \\ \lim_{n \rightarrow \infty} \frac{2(\frac{1}{3})^n + (\frac{2}{3})^n}{3(\frac{1}{3})^n + 1} = \frac{0+0}{0+1} = 0 \end{aligned}$$

~~G.T.H.P.~~

N.B

$$\frac{2 + 2^n}{3 + 3^n}$$

$$< \frac{2 + 2^n}{3^n}$$

$$\sum_{n=1}^{\infty} \left[\frac{2}{3^n} + \frac{2^n}{3^n} \right]$$

$$\sum_{n=1}^{\infty} 2\left(\frac{1}{3}\right)^n + \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$$

$$r = \frac{1}{3}$$

$$|r| < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} 2\left(\frac{1}{3}\right)^n$ is Convergent (Geometric Series)

$$r = \frac{2}{3}$$

$$|r| < 1$$

$\Rightarrow \therefore \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n$ is Convergent (Geometric Series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \left[\frac{2}{3^n} + \frac{2^n}{3^n} \right]$ is Convergent

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{2 + 2^n}{3 + 3^n}$ is Convergent (B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{1}{n^4 + n^2 + 5}$$



Solution #1



$$\frac{1}{n^4 + n^2 + 5} < \frac{1}{n^4 + n^2} < \frac{1}{n^4}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^4}$$

$$p = 4 > 1$$

Convergent (p-series)

$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1}{n^4 + n^2 + 5}$
is convergent

(B.C.T)



Solution #2



$$\frac{1}{n^4 + n^2 + 5} < \frac{1}{n^4 + n^2} < \frac{1}{n^2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

$$p = 2 > 1$$

Convergent (p-series)

$$\Rightarrow \therefore \sum_{n=1}^{\infty} \frac{1}{n^4 + n^2 + 5}$$

is convergent

(B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=2}^{\infty} \frac{\ln(n)}{n^5}$$

Let $f(x) = \frac{\ln x}{x^5}$

~~G.T.H.P.I~~ ✓

1) $f(x)$ is continuous and positive on $[2, \infty)$

N.B

2) $f'(x) = \frac{x^5 \cdot \frac{1}{x} - \ln x \cdot 5x^4}{(x^5)^2}$

$$= \frac{x^4 - \ln x \cdot 5x^4}{x^{10}} = \frac{x^4 [1 - 5\ln x]}{x^{10}}$$

$$= \frac{1 - 5\ln x}{x^6} < 0 \quad \forall x \in [2, \infty)$$

$\therefore f(x)$ is decreasing on $[2, \infty)$

$$\int_2^{\infty} \frac{\ln x}{x^5} dx$$

$$I_1 = \int \frac{\ln x}{x^5} dx$$

$$u = \ln x \quad dv = \frac{1}{x^5} dx$$

$$du = \frac{1}{x} dx \quad v = x^{-4} \quad \text{(with } x^{-5} \text{ circled in the original image)}$$

$$= \frac{1}{4} \ln x \cdot x^{-4} + \frac{1}{4} \int x^{-4} \cdot \frac{1}{x} dx$$

$$= \frac{1}{4} \ln x \cdot \frac{1}{x^4} + \frac{1}{4} \frac{x^{-4}}{-4} + C$$

$$= \frac{1}{4} \frac{\ln x}{x^4} - \frac{1}{16} \frac{1}{x^4} + C$$



See next page plz!

$$\lim_{t \rightarrow \infty} \int_2^t \frac{\ln x}{x^5} dx$$

$$\lim_{t \rightarrow \infty} \left[-\frac{1}{4} \frac{\ln x}{x^4} - \frac{1}{16} \frac{1}{x^4} \right]_2^t$$

$$\lim_{t \rightarrow \infty} \left[\left(-\frac{1}{4} \frac{\ln t}{t^4} - \frac{1}{16} \frac{1}{t^4} \right) - \left(-\frac{1}{4} \frac{\ln 2}{(2)^4} - \frac{1}{16} \frac{1}{(2)^4} \right) \right]$$

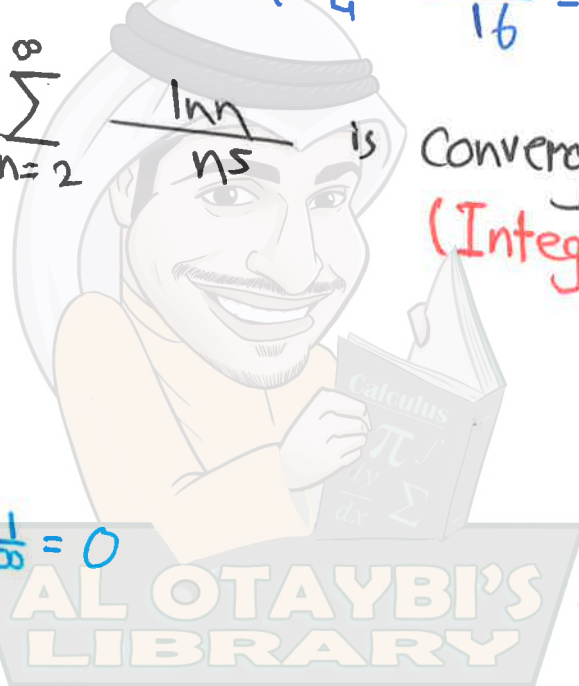
$$= (0 - 0) - \left(-\frac{1}{4} \frac{\ln 2}{16} - \frac{1}{16} \frac{1}{16} \right)$$

$\Rightarrow \therefore \sum_{n=2}^{\infty} \frac{\ln n}{n^5}$ is convergent
(Integral test)

$$\lim_{t \rightarrow \infty} \frac{\ln t}{t} \quad \frac{\infty}{\infty}$$

L'Hop

$$\lim_{t \rightarrow \infty} \frac{\frac{1}{t}}{1} = \frac{1}{\infty} = 0$$



Determine whether the following series is convergent or divergent

$$\sum_{n=2}^{\infty} \frac{\ln(n)}{n^5}$$

$\ln n < n \quad \forall n$
 $\div n^5$

$\frac{\ln n}{n^5} < \frac{n}{n^5}$

$\frac{\ln n}{n^5} < \frac{1}{n^4}$

$\sum_{n=2}^{\infty} \frac{1}{n^4} \quad p=4 > 1$

Convergent (P-series)

$\Rightarrow \therefore \sum_{n=2}^{\infty} \frac{\ln n}{n^5}$ is convergent (B.C.T)

~~G.T.H.P.I~~

~~N.B~~

Determine whether the following series is convergent or divergent

$$\sum_{n=3}^{\infty} \frac{\ln(n)}{n^{\frac{3}{4}}}$$

G.T.H.P.I

N.B

لاحظ أن البداية 3
هناك علاقتين لـ $\ln$



$$\ln n > 1 \quad \forall n > 3 \quad \div n^{\frac{3}{4}}$$

$$\frac{\ln n}{n^{\frac{3}{4}}} > \frac{1}{n^{\frac{3}{4}}}$$

$$\sum_{n=3}^{\infty} \frac{1}{n^{\frac{3}{4}}} \quad P = \frac{3}{4} < 1$$

is divergent (P-series)

$$\Rightarrow \therefore \sum_{n=3}^{\infty} \frac{\ln n}{n^5} \text{ is divergent}$$

(B.C.T)

$$\ln n < n \quad \forall n > 0 \quad \div n^{\frac{3}{4}}$$

$$\frac{\ln n}{n^{\frac{3}{4}}} < \frac{n}{n^{\frac{3}{4}}} = \frac{1}{n^{\frac{1}{4}}}$$

$$\frac{\ln n}{n^{\frac{3}{4}}} < \frac{1}{n^{-\frac{1}{4}}}$$

$$\sum_{n=3}^{\infty} \frac{1}{n^{-\frac{1}{4}}} \quad P = -\frac{1}{4} < 1$$

is divergent

* فشل الاختبار *



Text book questions



Determine whether the series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{9^n}{3 + 10^n} \rightarrow a_n$$

$$0 < \frac{9^n}{3 + 10^n} < \frac{9^n}{10^n}$$

$$\sum_{n=1}^{\infty} \frac{9^n}{10^n} = \sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$$

$$r = \frac{9}{10}$$

$$\therefore |r| < 1$$

$\therefore \sum_{n=1}^{\infty} \left(\frac{9}{10}\right)^n$ is convergent
(Geometric Series)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)

Determine whether the series converges or diverges.

$$\sum_{n=1}^{\infty} \frac{\arctan n}{n^{1.2}} \rightarrow a_n$$

$$\frac{\pi}{4} < \tan^{-1} n < \frac{\pi}{2} \quad \forall n \geq 1$$

$$\frac{\frac{\pi}{4}}{n^{1.2}} < \frac{\tan^{-1} n}{n^{1.2}} < \frac{\frac{\pi}{2}}{n^{1.2}}$$

$$\sum \frac{\frac{\pi}{2}}{n^{1.2}} \quad p = 1.2 > 1$$

is convergent (P-series)

$\therefore \sum_{n=1}^{\infty} a_n$ is convergent (B.C.T)

Determine whether the following series is convergent or divergent

$$\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n}$$

$$\frac{\pi}{4} < \tan^{-1} n < \frac{\pi}{2}$$

~~G.T.H.P.I~~

~~N.B~~

$$\frac{\pi}{4} < \frac{\tan^{-1} n}{n} < \frac{\frac{\pi}{2}}{n}$$

is divergent (Harmonic)

$$\therefore \sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n} \text{ is divergent (B.C.T)}$$

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Determine whether the following series is convergent or divergent

$$\sum_{n=3}^{\infty} \frac{\ln(n)}{\sqrt{n} \sin^2 n}$$



Solution #1



$$0 < \sin^2 n < 1$$

$$\ast \sqrt{n}$$

$$0 < \sqrt{n} \sin^2 n < \sqrt{n}$$

$$\sqrt{n} \sin^2 n < \sqrt{n}$$

$$\frac{1}{\sqrt{n} \sin^2 n} > \frac{1}{\sqrt{n}}$$

$$\therefore \ln n > 0 \quad \forall n > 1$$

$$\ast \ln n$$

$$\frac{\ln n}{\sqrt{n} \sin^2 n} > \frac{\ln n}{\sqrt{n}} > \frac{1}{\sqrt{n}}$$

$$\sum_{n=3}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=3}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad p = \frac{1}{2} < 1$$

divergent (P-series)

$$\Rightarrow \therefore \sum_{n=3}^{\infty} \frac{\ln(n)}{\sqrt{n} \sin^2 n} \text{ is divergent}$$

(B.C.T)



Solution #2



$$\ln n > 1 \quad \forall n$$

$$\div \sqrt{n} \sin^2 n$$

$$\frac{\ln n}{\sqrt{n} \sin^2 n} > \frac{1}{\sqrt{n} \sin^2 n} > \frac{1}{\sqrt{n}}$$

$$\sum_{n=3}^{\infty} \frac{1}{\sqrt{n}} = \sum_{n=3}^{\infty} \frac{1}{n^{\frac{1}{2}}} \quad p = \frac{1}{2} < 1$$

divergent (P-series)

$$\Rightarrow \therefore \sum_{n=3}^{\infty} \frac{\ln(n)}{\sqrt{n} \sin^2 n}$$

is divergent
(B.C.T)